ORBITAL 26 · ARTEMIS · TEAM ATLAS

ATLAS

AR CAMPUS ASSISTANT

Context-aware wayfinding from campus routes to indoor facilities.

Overview

Atlas helps NUS students move beyond ordinary map search. It combines campus points, indoor facilities, schedules, and recommendation logic so the app can suggest where to go next and what facility is useful nearby.

Core idea

- Outdoor route
- Indoor facility
- Schedule context
- Daily guidance

Features

1. **AR Wayfinding**
   Route lines, waypoints, and future AR overlays guide students through buildings.
2. **Facility Discovery**
   Find study spaces, restrooms, lifts, and printing points by building and type.
3. **Daily Agent**
   Schedule-aware recommendations surface the next useful campus action.
4. **NUSMods Sync**
   External module data integration keeps the system extensible.
5. **Protected Demo Flow**
   Login and JWT-protected routes model a realistic user journey.

System Flow

Campus Data → Auth + API → Schedule Context → Smart Guidance

Testing

API	health, login, JWT protection, schedule CRUD
Build	Vite production build and Go package checks
Data	seeded campus scenarios for repeatable demos

Design Principles

1. **Context before controls**
   Recommend the next useful action instead of forcing search first.
2. **Indoor continuity**
   Campus navigation continues after the user reaches a building.
3. **Fast scanning**
   Dense but clear cards support repeated daily use.
4. **Progressive AR**
   Start from reliable map data, then layer AR visuals and cues.

SWE Practice

- **Layered architecture**
  React UI, API handlers, store layer, scheduler, client modules.
- **REST contracts**
  JSON routes for buildings, facilities, schedule, recs, auth, sync.
- **Security baseline**
  Bcrypt password hashing and JWT-protected workflows.
- **Reproducible demo**
  SQLite migration and seed data for stable local runs.

Tech Stack

React · Vite · Go API · SQLite · JWT · NUSMods

TEAM MEMBERS

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